

Internal proof audit for the degree-3 spectral refinement

Repository source: proof/PROOF_AUDIT.md

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Initial audit date: 2026-09-02

Author-review update: 2026-09-05

Release-candidate update: 2026-09-06

Scope: proof/BRIDGE_LEMMAS.md, proof/DEGREE3_TENSOR_IDENTITY.md, the exact Python checks, the independent R checks, and the current integrated LaTeX manuscript.

Audit conclusion

No internal algebraic or logical contradiction was found in the proposed proof chain. The bridge lemmas, cubic tensor identity, and final coefficient agree with the exact symbolic calculations and the numerical cross-checks.

This conclusion is deliberately limited:

- it is an internal project audit, not independent expert review;
- it does not establish novelty against all existing literature;
- it does not verify the stronger HYRA manuscript;
- it does not justify describing the result as peer reviewed or certified.

The appropriate public status remains:

AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification.

1. Source audit

The inherited statements used in the proof were checked against the published version of Nishioka's paper.

ID	Inherited statement	Source location	Audit result
SRC-1	For the centered support function, $h(-u) = -h(u)$ and $A_h = \nabla^2 h + h I$ satisfies $-dI/2 \leq A_h \leq dI/2$	Nishioka, Section 2.1, equation (3)	Matches project notation after identifying tangent spaces with $u^\perp$
SRC-2	$\text{Vol}(K) = \pi d^{3/6} - d E(h)/2$	Nishioka, Lemma 1, equation (4)	Matches exactly
SRC-3	Degree-one harmonics represent translations and may be removed	Nishioka, Section 2.3	Used correctly; A_h and $E(h)$ are unchanged by the removed translation mode
SRC-4	$E(Y_1)$ and $\ A_{Y_1}\ _2^2$ have the spectral factors used here	Nishioka, Lemma 4	Matches exactly
SRC-5	Distinct harmonic degrees contribute orthogonally	Nishioka, proof of Proposition 1; polarized explicitly in this project	Matches; the project supplies the tensor-bilinear version needed for the projection identity
SRC-6	Bodies of constant width admit same-width smooth approximation and volume is Hausdorff-continuous	Nishioka, Lemma 2; Martini–Montejano–Oliveros, Theorem 8.5.1 for analytic-boundary approximation	Sufficient for passage from the smooth proof to arbitrary bodies

2. Bridge-lemma audit

BR-1: normalization and translation

The paper now distinguishes two operations that were easy to conflate:

1. subtracting $d/2$ from the support function, which produces the odd centered function h ; and
2. translating the body to eliminate the degree-one component.

Both operations preserve the width and volume. The degree-one component also lies in the kernel of $f \rightarrow A_f$, so the curvature estimates are unaffected.

Status: checked.

BR-2: tensor-valued harmonic orthogonality

Polarizing the integrated Bochner identity gives

$$\int \langle \text{Hess } f, \text{Hess } g \rangle = \int (\Delta f)(\Delta g) - \int \langle \text{grad } f, \text{grad } g \rangle.$$

Expanding $\langle A_f, A_g \rangle$ then reduces every term to a scalar multiple of $\int f g$ when f and g are spherical harmonics. Distinct degrees are therefore orthogonal for the tensor quadratic form. Consequently,

$$\|A_h\|_2^2 = \sum_1 \|A_{h_1}\|_2^2, \\ \|A_{h_3}\|_2^2 = \langle A_h, A_{h_3} \rangle_2.$$

The convergence step is justified because a smooth h has harmonic expansion converging in $H^2(S^2)$, while $f \rightarrow \text{Hess } f + f I$ is continuous from H^2 to tensor-valued L^2 .

Status: checked.

BR-3: refined spectral split

For odd degrees,

$$E(h_1) = ||A_{h_1}||_2^2 / (2(1(1+1)-1)).$$

The degree-three coefficient is $1/22$; the largest coefficient among all remaining odd degrees is the degree-five coefficient $1/58$. Therefore

$$E(h) \leq ||A_h||_2^2/58 + (9/319)||A_{h_3}||_2^2.$$

The arithmetic $1/22 - 1/58 = 9/319$ was checked exactly.

Status: checked.

BR-4: nuclear-norm identity and determinant scaling

For tangent eigenvalues μ_1, μ_2 ,

$$||C||_*^2 = |C|_F^2 + 2|\det C|.$$

With $q = 4 \det C$, this becomes

$$||C||_*^2 = |C|_F^2 + |q|/2.$$

The factor $1/2$ is correct and is carried consistently through both Cauchy–Schwarz applications.

Status: checked.

BR-5: operator/nuclear duality and the cubic budget

The projection identity gives a positive scalar exactly:

$$N_3 = \text{integral } \text{tr}(A_h C).$$

The pointwise box constraint implies $||A_h||_{\text{op}} \leq d/2$, and the standard dual norm inequality gives

$$N_3 \leq (d/2) \text{integral } ||C||_*.$$

Combining this with the determinant estimate yields

$$N_3 \leq \pi d^2 (1 + \sqrt{15183/17303}).$$

No sign assumption on $\text{tr}(A_h C)$ is used pointwise; the proof inserts an absolute value before applying the norm inequality.

Status: checked.

BR-6: final assembly

The global pointwise box gives $||A_h||_2^2 \leq 2 \pi d^2$. Substitution into the refined spectral split produces

$$E(h) \leq \pi d^2(20 + 9s)/319, \\ s = \sqrt{15183/17303}.$$

The volume identity then gives

$$\text{Vol}(K) \geq \pi(259 - 27s)d^3/1914.$$

The exact difference from Nishioka's coefficient is

$$9 \pi(1-s)/638 > 0.$$

Status: exact-code checked and hand-checked.

3. Degree-three tensor audit

T-1: harmonic tensor representation

For $F(x)=T_{\{ijk\}}x_i x_j x_k$,

$$\Delta F = 6 T_{\{iik\}} x_k.$$

Thus the restriction is a degree-three spherical harmonic exactly when the symmetric tensor is trace-free.

Status: checked symbolically and by hand.

T-2: spherical Hessian and curvature tensor

For tangent vectors, the restriction formula

$$\text{Hess}_S Y = D^2 F|_T - 3Y I$$

follows from the second fundamental form of the unit sphere and Euler's identity $DF(u)[u]=3F(u)$. Since $D^2F=6M$,

$$C = A_Y = 6PMP - 2YP.$$

The sign and the homogeneity factor 3 were checked against the trace identity $\text{tr } C = \Delta Y + 2Y = -10Y$.

Status: checked.

T-3: compact determinant formula

Using a basis with first vector u , the proof obtains

$$\begin{aligned} |PMP|^2 &= |M|^2 - 2|Mu|^2 + Y^2, \\ |C|^2 &= 36|M|^2 - 72|Mu|^2 + 68Y^2, \\ \text{tr } C &= -10Y. \end{aligned}$$

For a tangent 2×2 endomorphism represented by a 3×3 extension with zero normal eigenvalue,

$$4 \det C = 2((\text{tr } C)^2 - \text{tr}(C^2)).$$

This yields

$$4 \det C = 64Y^2 + 144|T(u,u,\cdot)|^2 - 72|T(u,\cdot,\cdot)|^2.$$

The direct projected-matrix formula and this compact formula agree exactly on $x^2+y^2+z^2=1$.

Status: exact-code checked by two formulations; hand derivation checked.

T-4: quadratic moments

The sphere moment normalization

$$\begin{aligned} &\text{integral } u_{\{i1\}} \dots u_{\{i,2m\}} \\ &= 4 \pi / (2m+1)!! \text{ times the sum over pairings} \end{aligned}$$

produces

$$\begin{aligned} \text{integral } Y^2 &= 8 \pi \tau / 35, \\ \text{integral } |v|^2 &= 8 \pi \tau / 15, \\ \text{integral } |M|^2 &= 4 \pi \tau / 3. \end{aligned}$$

These give $||C||_2^2 = 176 \pi \tau / 7$.

Status: checked by exact polynomial integration and pairing counts.

T-5: dimension-three quartic relation

Writing $(A_i)_{\{jk\}} = T_{\{ijk\}}$, the Cayley–Hamilton identity for traceless 3×3 matrices gives, after polarization and summation,

$$K_{\text{tet}} = \tau^2/2 - \text{tr}(B^2).$$

The index identification

$$\sum_{\{i,j\}} \text{tr}(A_i A_j A_i A_j) = K_{\text{tet}}$$

was expanded explicitly in the LaTeX draft and checked by the general symbolic tensor script.

Status: checked.

T-6: quartic moment table

All perfect pairings for the six moments were exhaustively enumerated. The counts reduce to the two invariants τ^2 and $J = \text{tr}(B^2)$ using the quartic relation. Substitution into the square of the compact determinant formula gives

$$\begin{aligned} \text{integral } (4 \det C)^2 \\ &= \pi(1310464 \tau^2 - 2265600 J)/1001 \\ &= \pi(555264 \tau^2 - 2265600 |B_0|^2)/1001. \end{aligned}$$

The exact upper bound follows by dropping the nonpositive B_0 term and using $||C||_2^2 = 176 \pi \tau/7$.

Status: exact-code checked; human-readable derivation drafted.

T-7: sharpness within the cubic harmonic space

For $\mathbf{Y} = \mathbf{xyz}$, the six symmetric components $T_{\{123\}}$ equal $1/6$, giving

$$\begin{aligned} \tau &= 1/6, \\ B &= I/18 = (\tau/3)I, \\ B_0 &= 0. \end{aligned}$$

Thus equality is attained in the determinant inequality, so its constant is sharp on H_3 .

Status: checked exactly.

4. Publication-draft audit

The current integrated manuscript is in:

```
paper/degree3_spectral_refinement.tex
paper/degree3_spectral_refinement.pdf
```

The PDF was built with `latexmk`, rendered at 180 dpi, and visually inspected page by page. The audit found no clipped text, overlapping equations, broken glyphs, or unresolved references. The release-candidate PDF contains the version and release date in its visible front matter and metadata. It has ten pages; the release build and PDF preflight confirm valid structure, extractable text, and embedded fonts.

The author-review revision adds a proof roadmap, an explicit derivation of the normalization in equation (14), a degree-five spectral-splitting remark, a discussion of the resulting spectral hierarchy, a representative mixed-moment pairing calculation, and stronger navigation to the expanded proof documents. These changes do not alter the claimed coefficient or the logical dependencies of the proof.

The manuscript makes the following provenance distinctions explicit:

- Nishioka’s support-function and spectral framework is sourced;
- the determinant inequality and cubic curvature budget are the proposed new contribution;
- HYRA is acknowledged as a public manuscript claiming stronger numerical bounds by a different method;
- exact symbolic checks are distinguished from independent expert review;
- AI assistance is disclosed by system and access period, and no AI system is listed as an author.

5. Literature-search status

A targeted exact-phrase and exact-constant web search on 2026-09-02 did not locate an earlier public occurrence of either 15183/17303 or the decimal coefficient 0.3836027047 in connection with the three-dimensional Blaschke–Lebesgue problem. The search did locate Nishioka’s 2026 arXiv record and unrelated spherical Blaschke–Lebesgue work.

This was not a systematic literature review and must not be treated as a proof of novelty. A specialist may know relevant representation-theoretic or invariant inequalities under different notation.

6. Fresh-clone verification and release-candidate status

A separate clone of commit `96cea57275070f21babf30477ab1b128ec0f5eb6` successfully ran the exact Python suite, the independent R suite, the paper build, the Markdown-PDF build, and the metadata and checksum checks. The rebuilt tracked working tree remained byte-clean. Full Poppler PDF preflight was unavailable on the local Mac and was delegated to the green GitHub Actions document job.

A second fresh-clone smoke test on release-candidate commit `ff5118cc05ada29d26631c82d0cbf69613ed88c1` passed the exact Python suite, the independent R suite, both document builds, release and metadata preflights, basic PDF validation, and checksum verification. The strict binary comparison found only two-byte differences in two secondary rendered documentation PDFs, with corresponding checksum updates. No source, proof statement, release metadata, or mathematical output changed. Exact PDF byte identity is therefore not treated as a scientific or release-blocking criterion. The detailed records are in `CLEAN_CLONE_CHECK.md`.

The named author has completed the structured understanding review, read and approved the revised manuscript and supporting package, and authorized preparation of release `v0.1.0`. The release version, release date, canonical repository URL, versioned release URL, public README wording, citation metadata, and release notes are fixed in the release-candidate files.

The remaining internal gate is procedural: the reviewed commit and the `v0.1.0` tag must each obtain three green hosted jobs. The tagged commit and the verification scope are recorded on the GitHub release page.

The exact Python and R checks, hosted three-job GitHub Actions workflow, research-paper build, Markdown-PDF build, metadata declarations, PDF preflight, and fresh-clone smoke tests do not replace independent subject-matter review.

Release recommendation

Keep the repository private through the reviewed commit and hosted checks. Once the exact commit has three green `main` jobs, tag it `v0.1.0`; require the three tag jobs to pass, then make the repository public and publish the GitHub release using the status language preserved throughout this package. Independent mathematical review remains the principal scientific next step.